

STARTER



These three cups have been filled with water.

Would you expect these three cups to have:

- a** exactly the same mass
- b** masses that were nearly the same
- c** masses that were very different?

Discuss this with a partner. Give reasons for your choice.

3:00

Structure of Atom

Lesson 1

Learning Objectives

1. Describe the structure of the atom
2. List the particles found in an atom
3. Describe some of the properties of the particles found in an atom

Success criteria

Lesson Title	Working towards	Working at	Working beyond
The structure of the atom	I can list the sub-atomic particles found in an atom.	I can describe the structure of an atom.	I can describe discoveries that led to the atomic model we use today.
Purity	I can describe what an element is.	I can explain what purity means.	I can calculate and interpret percentage purity.
Filtration	I can follow steps to filter a mixture.	I can describe the purpose and parts of the filtration setup.	I can explain why filtration separates substances using particle size.
Simple Distillation	I can describe what distillation does.	I can label a simple distillation setup.	I can explain how boiling and condensation separate substances.
Crystallisation	I can describe how crystals can form.	I can carry out a crystallisation method safely.	I can explain how solubility and evaporation lead to crystal formation.
Chromatography	I can state that coloured substances can be separated using chromatography.	I can describe and carry out a chromatography experiment.	I can analyse and explain chromatograms to identify substances.

**JUST HOW
SMALL
IS AN ATOM?**



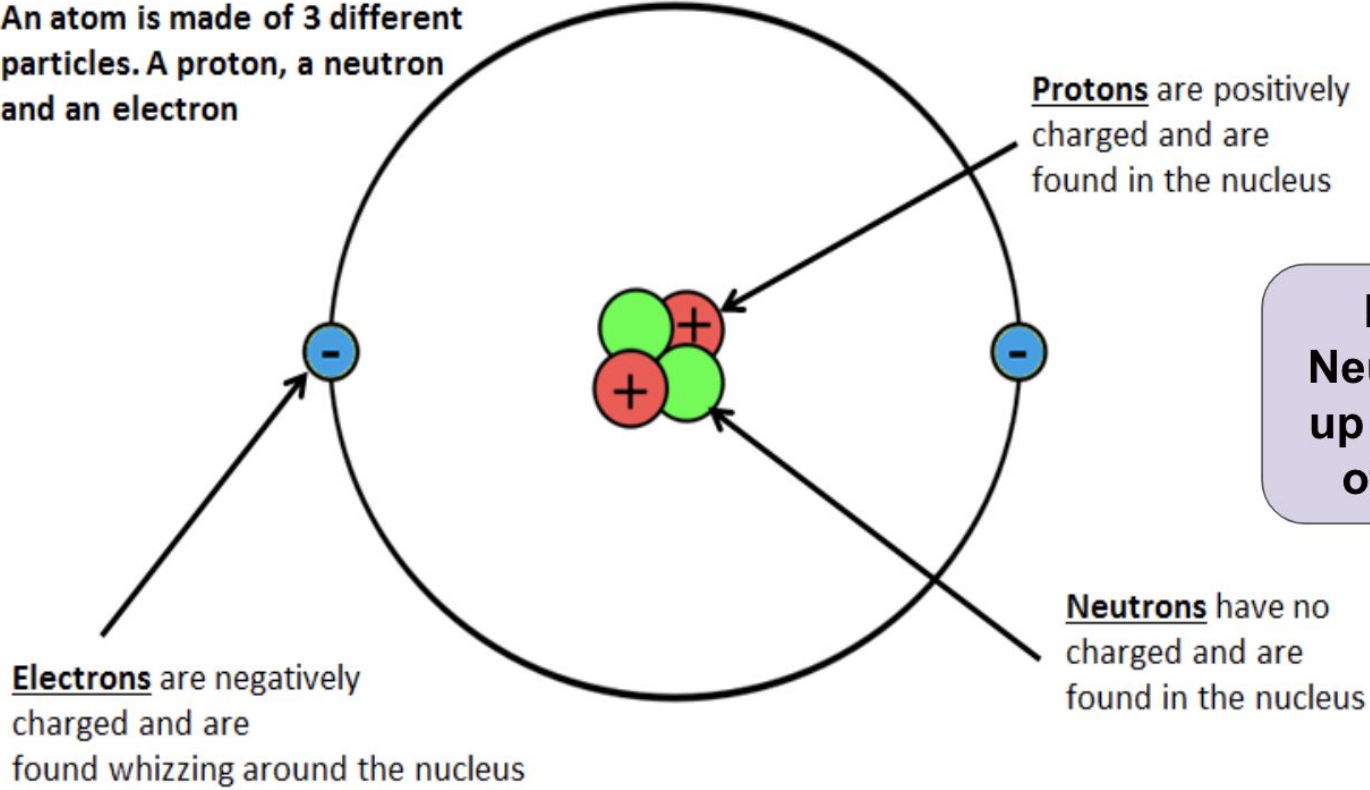
What is an Atom?



- **Atom** – Comes from the Greek word “*Atomos*” – Meaning **Indivisible**.
- This means it cannot be divided or split up anymore!
- So the smallest part of any substance is called an **atom**.



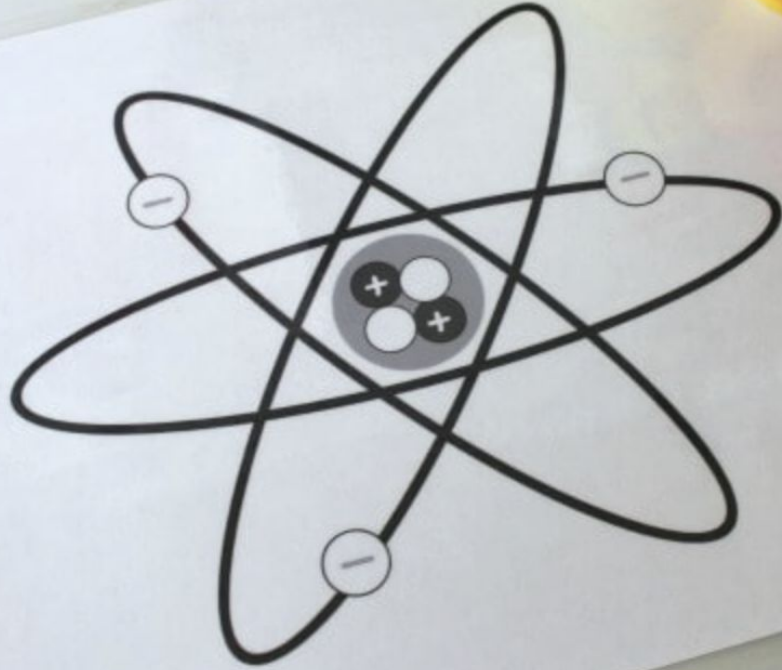
An atom is made of 3 different particles. A proton, a neutron and an electron



Protons & Neutrons make up the Nucleus of the atom!

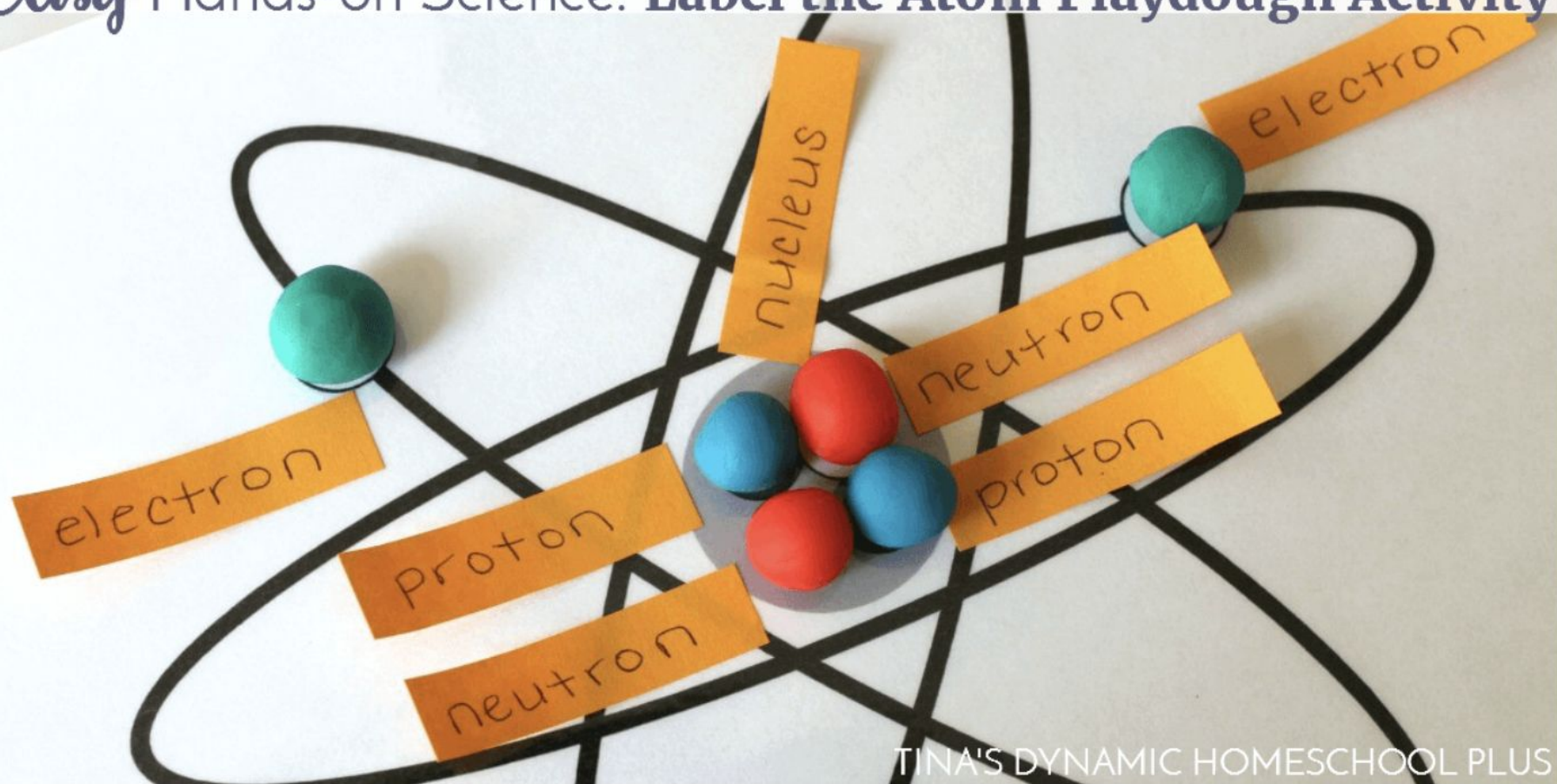
10:00

10:00



Use playdough (one colour for each subatomic particles) to create your atom model - use sticky notes to complete your diagram

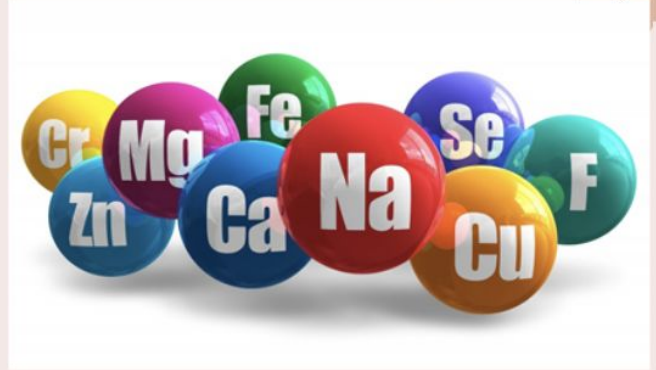
Easy Hands-on Science: Label the Atom Playdough Activity



Atoms & Elements



- There are **different** types of **atoms**.

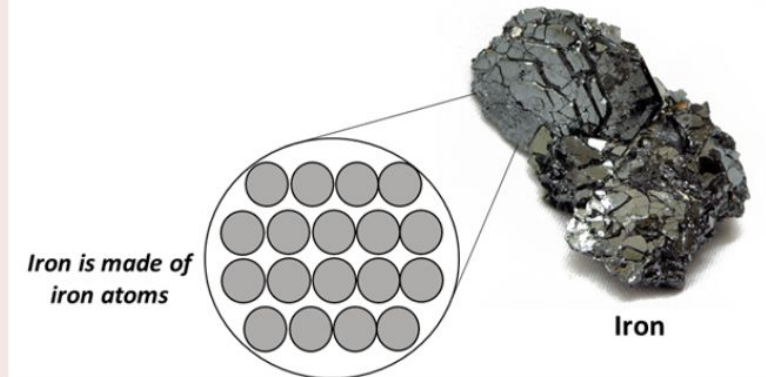
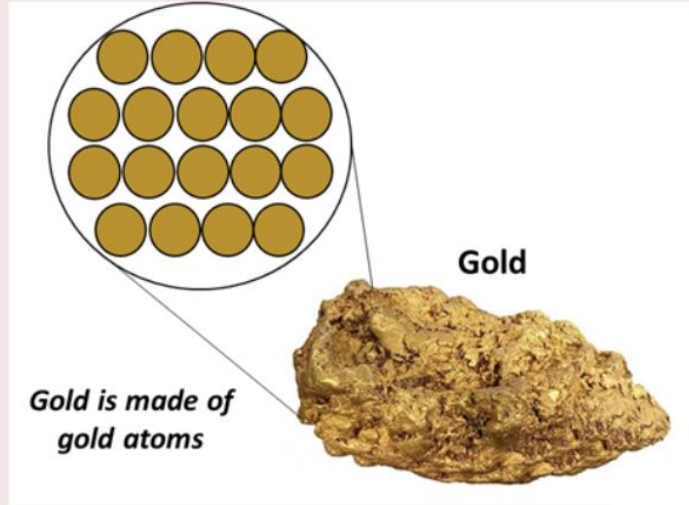


- Atoms are the smallest parts of pure substances called **elements**.

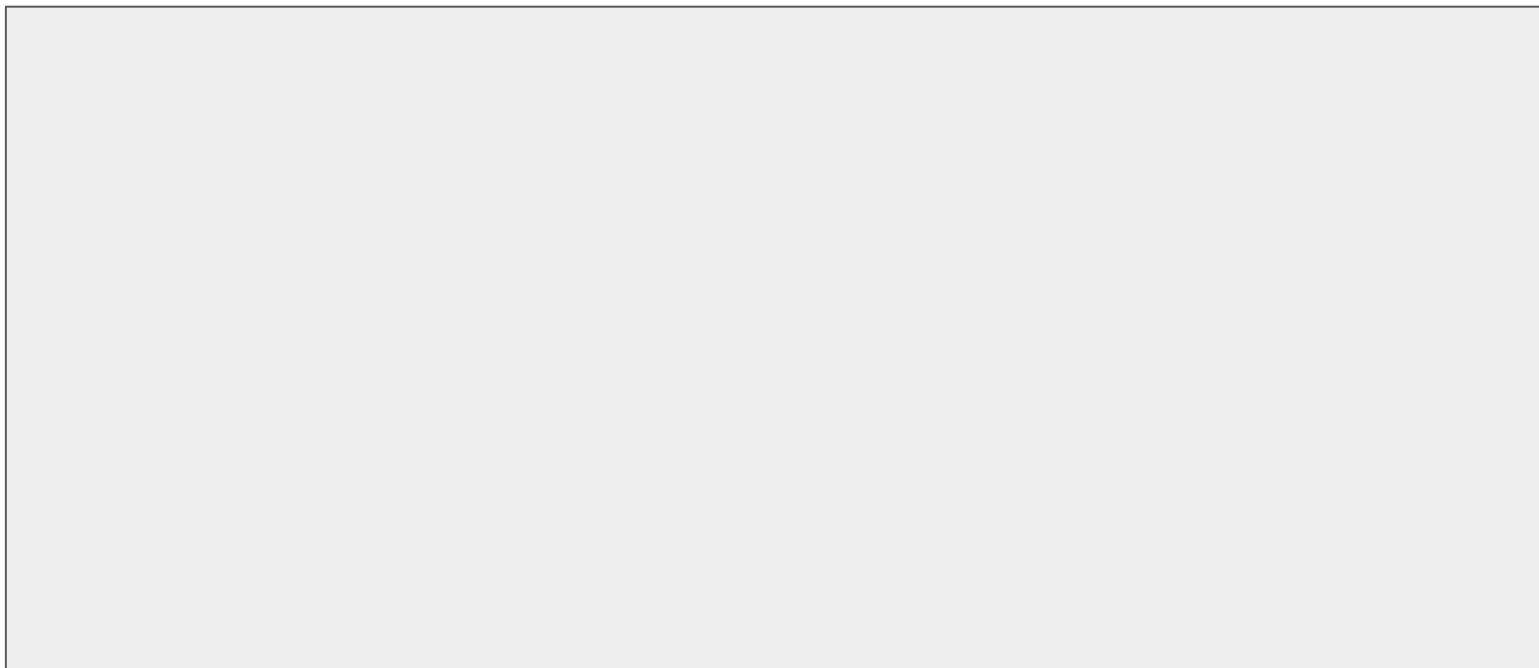
- **Element** = substance that is made up of just **1 type of atom**



Example of Elements



WHY DO SCIENTISTS USE SYMBOLS FOR ELEMENTS?



WHY DO SCIENTISTS USE SYMBOLS FOR ELEMENTS?

1. Using symbols is the easiest way to represent the elements
2. Use symbols is to allow us to write chemical formula easily.



DMITRI MENDELEEV

1. Created the **PERIODIC TABLE**
2. Arranged elements in order of **increasing atomic mass**
3. Put elements with **similar properties** into **GROUPS**
4. Arranged elements into rows called **PERIODS**

PERIODIC TABLE



Elements are found in the Periodic Table

hydrogen 1 H 1.0079																	helium 2 He 4.0026	
lithium 3 Li 6.941	beryllium 4 Be 9.0122											boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180	
sodium 11 Na 22.990	magnesium 12 Mg 24.305											aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948	
potassium 19 K 39.098	calcium 20 Ca 40.078	scandium 21 Sc 44.956	titanium 22 Ti 47.867	vanadium 23 V 50.942	chromium 24 Cr 51.996	manganese 25 Mn 54.938	iron 26 Fe 55.845	cobalt 27 Co 58.933	nickel 28 Ni 58.693	copper 29 Cu 63.546	zinc 30 Zn 65.39	gallium 31 Ga 69.723	germanium 32 Ge 72.61	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80	
rubidium 37 Rb 85.468	strontium 38 Sr 87.62	yttrium 39 Y 88.906	zirconium 40 Zr 91.224	niobium 41 Nb 92.906	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	ruthenium 44 Ru 101.07	rhodium 45 Rh 102.91	palladium 46 Pd 106.42	silver 47 Ag 107.87	cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29	
caesium 55 Cs 132.91	barium 56 Ba 137.33	57-70 ★	lutetium 71 Lu 174.97	hafnium 72 Hf 178.49	tantalum 73 Ta 180.95	tungsten 74 W 183.84	rhenium 75 Re 186.21	osmium 76 Os 190.23	iridium 77 Ir 192.22	platinum 78 Pt 195.08	gold 79 Au 196.97	mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]
francium 87 Fr [223]	radium 88 Ra [226]	89-102 ★ ★	lawrencium 103 Lr [262]	rutherfordium 104 Rf [261]	dubnium 105 Db [262]	seaborgium 106 Sg [266]	bohrium 107 Bh [264]	hassium 108 Hs [269]	meitnerium 109 Mt [268]	unnilium 110 Uun [271]	ununium 111 Uuu [272]	unbibium 112 Uub [277]	ununquadium 114 Uuq [289]					

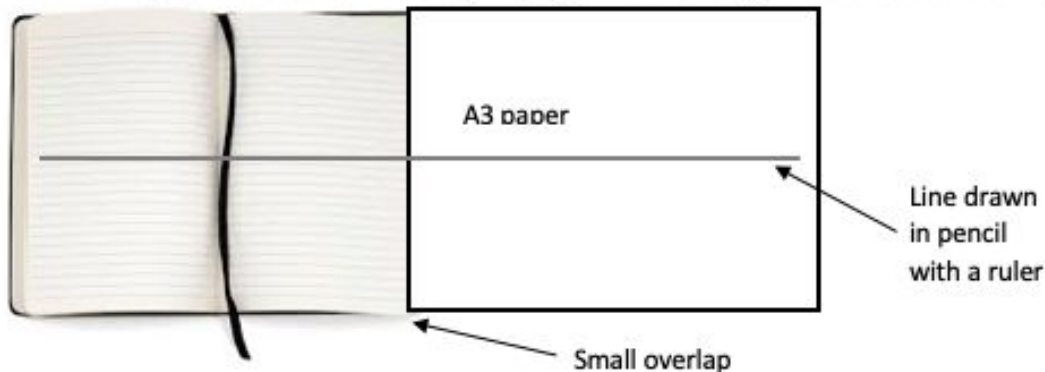
* Lanthanide series

** Actinide series

lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.04
actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]

Instructions

1. Open your book to a double blank page.
2. Orient the A3 paper landscape. Glue the left edge of the A3 to the right edge of your book. Overlap them just enough to ensure they stick.
3. Draw a horizontal line in pencil across the middle of your page. Afterwards, your book should look like this:



4. You may arrange your page landscape or portrait, but ensure you start on the lined paper of your book! The A3 can always be trimmed shorter.
5. Make the timeline: On the pencil line, draw the atomic model. Label the year model change (or publication leading to the model change) close to the line. The reason you drew this line in pencil is so that you can rub it out if it is going through your model. Above and below your pencil line, include information such as:
 - what the discovery was;
 - who made it (include his picture!); and
 - how the discovery was made (e.g. the experiment).

This is not an exercise in writing tiny to include every bit of detail. Make sure to include the important points and get rid of the “fluff”. Also ensure that your timeline is easy to read and appealing to study from—if you like looking at it, chances are you’ll look at it (and study it) more.

40:00

**Read &
Note-Taking!**

Examples of Student's Work

Atomic Theory Timeline



Democritus
(470-380 BC)

Greek Philosopher
Proposed that
matter could
not be divided
into smaller pieces
Forever; matter is
made from small
hard particles called
"atoms".

small
Hard
particle

Excellent timeline!
of how the culture



Charles-Augustin de
Coulomb (1736-1806)

-1785 Published work
on electrostatic
forces, saying op-
posite charges
attract and how
force is related
to charges and the
distance between
them



John Dalton
(1766-1844)

Confirmed that atoms 1808 -
are tiny, solid balls.
Created the first "atomic
theory". 1. Atoms are tiny in-
visible particles. 2. Atoms of an
element are all the same.
3. Atoms of different elements
are different. 4. Compounds are
formed by combining atoms.



J.J. Thomson
(1856-1940)

Negatively charged
Electron
1897
Discovered electrons
with cathode rays.
First scientist to
show the atom was
made of even
smaller things called
atomic matter.
To plum pudding model



Ernest Rutherford
(1871-1937)

Discovered protons and the
nucleus with the gold foil
experiment. Showed that
the atom has a positively
charged particle in the
center called a proton.
Committed that atoms are
mostly empty space.
He called the center
of the atom nucleus.



Hydrogen is a gas
Experiment
Rutherford's gold foil
experiment
Alpha particles went straight
through some scattered
a bit. Others scattered
back. Also, this pattern was
caused by the atom not
having a positive force
center.



Niels Bohr
(1885-1962)

1913 From thinking about
specific (energy) levels
arising from thinking
about Rutherford's model
proposed the electrons
had to move without
the nucleus in specific
layers of "shells". Every atom
has a specific number of
electron shells.



positively
charged
nucleus
Electrons
in shells
Rutherford - Bohr
or Bohr model

Lead to the idea
where electrons move
of being



Erwin Schrodinger
(1887-1961)

Published work
on quantum wave
equation and
the wave nature
of particles.



Named the
nucleus
protons
neutrons
Electron
clouds



Werner Heisenberg
(1901-1976)

Published the
uncertainty principle
are of his laws
states that
one can never
know the exact
location or
momentum
of an electron.



James Chadwick
(1891-1974)

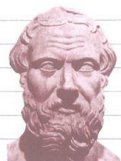
Discovered the neutron
by measuring its mass.
He even anticipated neu-
trons becoming a major be-
ar upon to fight cancer, but
arriving with Rutherford,
he discovered particles
with no charge and
could pass through
nuclei.

Examples of Student's Work

1785

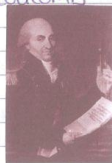
Publishes work on electrostatic forces, saying opposite charges attract and how force is related to charges and the distance between them

 Democritus (470 - 380 BC)



Democritus
(470 - 380 BC)

-400bc (1736-1806)
Charles-Augustin de Coulomb
all Hard
particle 'Atom'



1897

Discovered electrons with cathode rays. First Scientist to show the atom was made of even smaller things. He proposed model to "pudding" model.

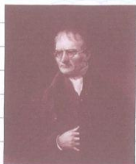


positively charged "banging"

negatively charged "marbles"

Plum-pudding model

John Dalton (1766-1844)



John Dalton
(1766-1844)

(1736-1806)
Charles-Augustin de Coulomb



Discovered electrons with cathode rays. First Scientist to show the atom was made of even smaller things. "Plum-charged model to pudding" model

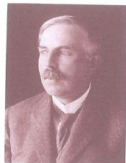
positively charged "dough"

negatively charged "raisins"

Plum- pudding model



Ernest Rutherford (1871-1937)



Ernest
Rutherford
(1871-1937)

1911
Positive charge
Rutherford
Nucleol
negative charges
Empty space
Nucleol

Discovered protons and the nucleus with the gold foil experiment. Showed that the atoms have +ve particles in the centre and are mostly empty space. He called the +ve 'protons' and the centre of the atom the



(1883-¹⁹⁶²~~1961~~)
Niels Bohr

A black and white portrait of a man with dark hair, wearing a suit, white shirt, and patterned tie. He is looking slightly to the right of the camera. The photo is mounted on a lined notebook page.

1913
From thinking about specific (energy) problems arising from thinking about 'models' of the atom, Rutherford proposed the electrons had to move around the nucleus in specific layers or 'shells'. Every atom has a specific number of electron shells.



Erwin
Schrödinger
(1887 - 1961)

A black and white portrait of a young man with dark hair, wearing a dark suit jacket, white shirt, and dark tie. He is looking slightly to the right of the camera with a neutral expression. The background is a plain, light-colored wall.

Werner
Heisenburg
(1901 - 1976)

Discovered the neutron by measuring its mass. Working with Rutherford he discovered particles with NO charge and called them neutrons, which were found in the nucleus with the protons.

(1891-1974)
James Chadwick



Electron
simultane-
ously:

59

Examples of Student's Work



Democritus
(470-380 BC)

Greek philosopher
Proposed that matter could not be divided into smaller pieces forever; matter is made from small, hard particles called "atoms."

Confirmed that atoms are tiny, solid balls. Created the first "atomic theory":
1. Atoms are tiny, invisible particles
2. Atoms of one element are all the same.
3. Atoms of different elements are different.
4. Compounds are formed by combining atoms.



John Dalton
(1766-1844)

Atomic Theory Timeline

400BC

small, hard particles

1785



Charles-Augustin de Coulomb
(1736-1806)

Publishes work on electrostatic forces, saying opposite charges attract and how force is related to charges and the distance between them.

1808

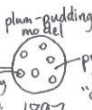


J. J. Thomson
(1856-1940)

model to

Discovered electrons with cathode rays. First scientist to show the atom was made of even smaller things. Changed atomic model to "plum pudding" model.

negatively charged "raisins"

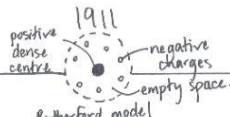


1897



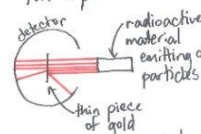
Ernest Rutherford
(1871-1937)

Discovered protons and the nucleus with the gold foil/gold leaf experiment. Showed that the atoms have positive particles in the central space. He called the positives "protons" and the centre of the atom the "nucleus".



Rutherford model

Rutherford's gold foil experiment



Most α particles went through the foil, some were scattered a little, but some were scattered A LOT! Concluded the atom must have a very dense, positive centre.



Erwin Schrödinger
(1887-1961)

Published work on (quantum) wave equation and treated atoms as waves instead of particles

Lead to the idea of electron "clouds" where electrons have a high probability of being



Werner Heisenberg
(1901-1976)

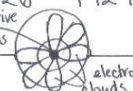
Published the "uncertainty principle". One of its forms states that one can never know the exact location of energy of an electron simultaneously.

1926

positive nucleus

1927

electron clouds



protons

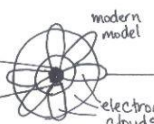
neutrons

1932



James Chadwick
(1891-1974)

Discovered the neutron by measuring its mass! (He even anticipated neutrons becoming a major weapon to fight cancer.) Working with Rutherford, he discovered particles with no charge and called them neutrons, which were found in the nucleus with the protons



modern model

electron clouds

???

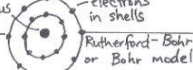


Niels Bohr
(1885-1962)

From thinking about specific (energy) problems arising from thinking about Rutherford's model, proposed the electrons had to move around the nucleus in specific layers or "shells". Every atom has a specific number of electron shells.

positive nucleus

electrons in shells



1913

From thinking about specific (energy) problems arising from thinking about Rutherford's model, proposed the electrons had to move around the nucleus in specific layers or "shells". Every atom has a specific number of electron shells.

Examples of Student's Work

Atomic Theory Timeline

400 BC

 Democritus (470-380 BC)
 Greek Philosopher
 Proposed that matter could not be divided into smaller pieces forever; matter is made from small, hard particles called atoms.

1785

 Charles-Augustin de Coulomb (1736-1806)
 Published work on electrostatic forces, saying opposite charges attract and have forces related to charges and the distance between them.

1808

 John Dalton (1766-1844)
 Confirmed that atoms are tiny solid balls, created the first atomic theory.
 1. Atoms are tiny invisible particles
 2. Atoms of one element are all the same
 3. Atoms of different elements are different
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1847

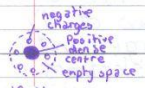
 Plum pudding model
 A positively charged dough with negative charged raisins.

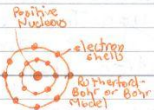
1869

 Dmitri Mendeleev (1834-1907)
 Discovered elements with periodic table. First scientist to show the atom was made of even smaller things. Changed atomic model to plum pudding model.

1897

 J.J. Thomson (1856-1940)
 Discovered electrons with cathode rays. First scientist to show the atom was made of even smaller things. Changed atomic model to plum pudding model.

1911

 Rutherford's Nuclear Model
 Positive nucleus, negative charges, empty space.

1913

 Bohr's Model
 Positive nucleus, electron shells, discrete energy levels.

1919

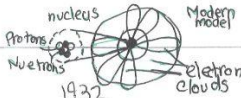
 Ernest Rutherford (1871-1937)
 Discovered Protons and the nucleus with the gold foil/gold leaf experiment. Showed that the atoms have positive particles in the centre and are mostly empty space. He called the positives "protons" and the centre of the atom nucleus.

1926

 Erwin Schrödinger (1887-1961)
 Published work on Quantum wave equation and treated atoms as waves instead of particles.

1927

 Heisenberg's Uncertainty Principle
 Lead to the idea of electron "clouds" where electrons have a probability of being.

1932

 Modern model
 nucleus, Protons, Neutrons, electron clouds.

1932

 James Chadwick (1891-1974)
 Discovered the neutron by measuring its mass! (He even anticipated neutrons becoming a major weapon to fight cancer.) Working with Rutherford, he discovered particles with no charge and called them neutrons which were found in the nucleus with the protons.

1927

 Werner Heisenberg (1901-1976)
 Published the "uncertainty principle". One of its forms states that one can never know the exact location/energy of an electron simultaneously.

<https://phet.colorado.edu/en/simulations/build-an-atom>

20:00



SIMULATIONS TEACHING RESEARCH INITIATIVES

DONATE



Build an Atom



Challenge

Questions

- 6 Explain how Thompson's model of the atom is different from the one scientists use today.
- 7 Who discovered the electron? When did they discover it?
- 8 Who proved that the neutron existed? When did they do this it?
- 9 What two things did Rutherford discover about the structure of the atom?
- 10 How is Rutherford's model different from the model scientists use today?